Note that from a given convergent sequence we can form not only two but a finite number m of new sequences converging to the same limit. One way to do it is as follows. The first new sequence will consist of the 1st, (m+1)st, (2m+1)st, (3m+1)st, etc. terms of the initial sequence. The second sequence will consist of the 2nd, (m+2)nd, (2m+2)nd, (3m+2)nd, etc. terms of the initial sequence. Similarly we can form the third, the fourth, and other sequences.

~~In conclusion, let us see how one can "spoil" a convergent~~

- In conclusion, let us see how one can "spoil" a convergent sequence by turning it into divergent. Clearly, different "spoiling" approaches are possible. Try to suggest something simple.

Reader: For example, we can replace all the terms with even serial numbers by a constant that is not equal to the limit of the initial sequence.

Sequence 5: $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}, \frac{8}{9}, \ldots\ldots$

Spoilt seq: $\frac{1}{2}, 2, \frac{3}{4}, 2, \frac{5}{6}, 2, \frac{7}{8}, 2, \ldots\ldots$

Author: I see that you have mastered very well the essence of the concept of a convergence sequence. Now we are ready for another substantial step, namely, consider one of the most important concepts in calculus, the defn. of a function.

Son: Resources for training in constructing and recognizing counterexamples

1. Counterexamples in Analysis — Gelbaum & Olmsted

This is the classic.

- Entire book dedicated to counterexamples
- Covers: 1) sequences 2) limits 3) continuity 4) integration
- Shows how statements fail
- Often gives multiple counterexamples for the same idea.

2. Counterexamples in Calculus — Sergiy Klymchuk

- Much more accessible than the above
- Focuses on calculus-level ideas
- Builds intuition step-by-step.